Example 6.5

Evaluate: (i) $\sin 30° + \cos 30°$ (ii) $\tan 60^\circ \cot 60^\circ$

(iii) $\dfrac{\tan 45°}{\tan 30° + \tan 60°}$

(iv) $\sin^2 45° + \cos^2 45°$

Note

(i) $(\sin\theta)^2$ is written as $\sin^2\theta = (\sin\theta)\times(\sin\theta)$

(ii) $(\sin\theta)^2$ is not written as $\sin\theta^2$, because it may mean as $\sin(\theta\times\theta)$.

Solution

(i) $\sin 30° + \cos 30° = \dfrac{1}{2} + \dfrac{\sqrt{3}}{2} = \dfrac{1+\sqrt{3}}{2}$

(ii) $\tan 60^\circ \cot 60^\circ = \sqrt{3} \times \dfrac{1}{\sqrt{3}} = 1$

(iii) $\dfrac{\tan 45°}{\tan 30° + \tan 60°} = \dfrac{1}{\frac{1}{\sqrt{3}} + \frac{\sqrt{3}}{1}} = \dfrac{1}{\frac{1+\left(\sqrt{3}\right)^2}{\sqrt{3}}} = \dfrac{1}{\frac{1+3}{\sqrt{3}}} = \dfrac{\sqrt{3}}{4}$

(iv) $\sin^2 45° + \cos^2 45° = \left(\dfrac{1}{\sqrt{2}}\right)^2 + \left(\dfrac{1}{\sqrt{2}}\right)^2 = \dfrac{1^2}{\left(\sqrt{2}\right)^2} + \dfrac{1^2}{\left(\sqrt{2}\right)^2} = \dfrac{1}{2} + \dfrac{1}{2} = 1$

Thinking Corner

The set of three numbers are called as Pythagorean triplets as they form the sides of a right angled triangle. For example,

(i) 3, 4, 5 (ii) 5, 12, 13 (iii) 7, 24, 25

Multiply each number in any of the above Pythagorean triplet by a non-zero constant. Verify whether each of the resultant set so obtained is also a Pythagorean triplet or not.

Example 6.6

Find the values of the following:

(i) $(\cos 0° + \sin 45° + \sin 30°)(\sin 90° - \cos 45° + \cos 60°)$

(ii) $\tan^2 60° - 2\tan^2 45° - \cot^2 30° + 2\sin^2 30° + \dfrac{3}{4}\operatorname{cosec}^2 45°$

Solution

(i) $(\cos 0° + \sin 45° + \sin 30°)(\sin 90° - \cos 45° + \cos 60°)$

$= \left[1 + \dfrac{1}{\sqrt{2}} + \dfrac{1}{2}\right]\left[1 - \dfrac{1}{\sqrt{2}} + \dfrac{1}{2}\right]$

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